CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The net force needed to give a body of mass 10 kg an acceleration of 3 m s^{-2} is: 1

(a) 13 N **(b)** 30 N **(c)** 3.33 N **(d)** 60 N

2. A net force of 30 N acts on a body of mass 6 kg. Its acceleration is: 1

(a) 180 m s^{-2} **(b)** 0.2 m s^{-2} **(c)** 10 m s^{-2} **(d)** 5 m s^{-2}

3. The weight of a 20 kg body is (g = 10 m s^{-2}): 1

(a) 400 N **(b)** 200 N **(c)** 20 N **(d)** 100 N

4. Two forces of 12 N and 5 N act on a block along the same straight line in the same direction. The magnitude of the net force is: 1

(a) 17 N **(b)** 12 N **(c)** 60 N **(d)** 7 N

5. A body of mass 2 kg initially at rest is pushed by a net force of 6 N for 4 s. Its final velocity is: 1

(a) 6 m s^{-1} **(b)** 12 m s^{-1} **(c)** 14 m s^{-1} **(d)** 24 m s^{-1}

6. Newton's first law is also called the law of: 1

(a) momentum change **(b)** gravitation **(c)** inertia **(d)** action

7. Which of these is a pair of action–reaction forces? 1

(a) Weight of a book and the normal force on it
(b) Gravity and friction on a car
(c) Two tug-of-war teams' pulls on a rope
(d) Paddle pushes water backwards; water pushes paddle forwards

8. Airbags in cars protect passengers by: 1

(a) making the car stop faster
(b) increasing the time over which the passenger stops
(c) reducing the car's mass
(d) increasing friction

9. The value of g near the Earth's surface is approximately: 1

(a) 98 m s^{-2} **(b)** 1.6 m s^{-2} **(c)** 6.67 m s^{-2} **(d)** 9.8 m s^{-2}

10. An astronaut in free space throws a tool away from her. She will: 1

(a) stay at rest **(b)** move in the opposite direction
(c) move in the same direction **(d)** spin without moving

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define one newton. 2

12. Find the force required to accelerate a 1500 kg car at 2 m s^{-2}. 2

13. State Newton's third law of motion. 2

14. Why do action and reaction not cancel each other? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A 25 kg block is pushed with 55 N against a maximum friction of 50 N. Find the acceleration and displacement in 2 s. 3

16. Two children of different masses sit on identical swings. For which child will a bigger force be needed for the same acceleration? Why? 3

17. Explain with two examples how Newton's second law helps in protecting from injury. 3

OR

A weightlifter holds a 30 kg barbell steady. Find the force she applies (g = 9.8 m s^{-2}). 3

18. Explain with a sketch-based example how a canoe moves forward. 3

OR

Using Newton's third law, explain how we are able to walk. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A sports car of mass 1500 kg moves east: 0 to 10 m s^{-1} in 5 s, then constant for 5 s, then to rest in 5 s. 5

(i) Find the force in the first 5 s.
(ii) Find the net force in the middle interval.
(iii) Find the force during braking.
(iv) Find the total distance.

OR

(i) Why do airbags and cricket fielders reduce force on impact? 5
(ii) A 0.15 kg ball at 30 m s^{-1} is stopped in 0.1 s; find the force.
(iii) Find the force if stopped in 0.01 s.

20. (i) How does a rocket lift off? 5
(ii) How is the same principle used by the Chandrayaan-3 lander?
(iii) Explain with the balloon activity.

OR

Case study: A 0.1 kg bullet leaves a 5 kg gun with a force of 2 N. 5
(i) What is the recoil force on the gun?
(ii) Find the bullet's acceleration.
(iii) Find the gun's acceleration.
(iv) Explain the difference.

END OF QUESTION PAPER — Total: 40 Marks